10.34 Process Analysis

10.34.1 Purpose

Process analysis assesses a process for its efficiency and effectiveness, as well as its ability to identify opportunities for change.

10.34.2 Description

Process analysis is used for various purposes including:

- recommending a more efficient or effective process,
- determining the gaps between the current and future state of a process,
- understanding factors to be included in a contract negotiation,
- understanding how data and technology are used in a process, and
- analyzing the impact of a pending change to a process.

A number of frameworks and methodologies exist that focus on process analysis and improvement methods, such as Six Sigma and Lean. Methods for process improvement include value stream mapping, statistical analysis and control, process simulation, benchmarking, and process frameworks.

Common changes made to processes in order to improve them include:

- reducing the time required to complete a task or tasks in the process,
- modifying interfaces or hand-offs between roles and organizational units to remove errors, including the reduction or elimination of bottlenecks,
- automating steps that are more routine or predictable, and
- increasing the degree of automation in the decision making required by the process.

When analyzing a process, business analysts look for:

- how the process adds or creates value for the organization,
- how the process aligns to organizational goals and strategy,
- to what degree the process is and needs to be efficient, effective, repeated, measured, controlled, used, and transparent, and
- how the requirements for a solution cover the future state process and its external stakeholders, including customers.

10.34.3 Elements

.1 Identify Gaps and Areas to Improve

Identifying gaps and areas to improve helps to identify what areas are in scope for analysis. Industry-specific models and process frameworks may be helpful in this

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regard. When identifying gaps and areas to improve, business analysts:

- identify gaps between the current and desired future state,
- identify what gaps and areas are value and non-value added,
- understand pain points and the challenges of the process from multiple points of view,
- understand opportunities to improve the process from multiple points of view,
- align the gaps and areas to improve with the strategic direction of the organization, and
- understand the relationship of the gaps and areas to improve to changes in the enterprise.

.2 Identify Root Cause

Identifying the root cause of the gaps and improvement areas ensures that the solution addresses the right gap and area.

When identifying the root cause, business analysts understand:

- there may be multiple root causes,
- the inputs leading to the gap or area of improvement,
- who the right people are to identify the root cause, and
- the current measurements and motivators in place for those owning or performing the process.

.3 Generate and Evaluate Options

Generating options and alternative solutions to solve for the gap or area of improvement helps the team evaluate and see different points of view for improving the process. It is important for stakeholders to be involved in identifying the impact, feasibility, and value of the proposed solution relative to alternative options.

.4 Common Methods

SIPOC

SIPOC is a process analysis method that originates in the Six Sigma methodology and has been more commonly adopted as a process analysis method outside of Six Sigma.

It is used to look at the process and understand the Suppliers, Inputs, Process, Outputs and Customers of the process being analyzed.

A SIPOC provides a simple overview of the process. It also shows the complexity of who and what is involved in creating inputs to the process and shows who receives outputs from the process. A SIPOC is a powerful tool used to create

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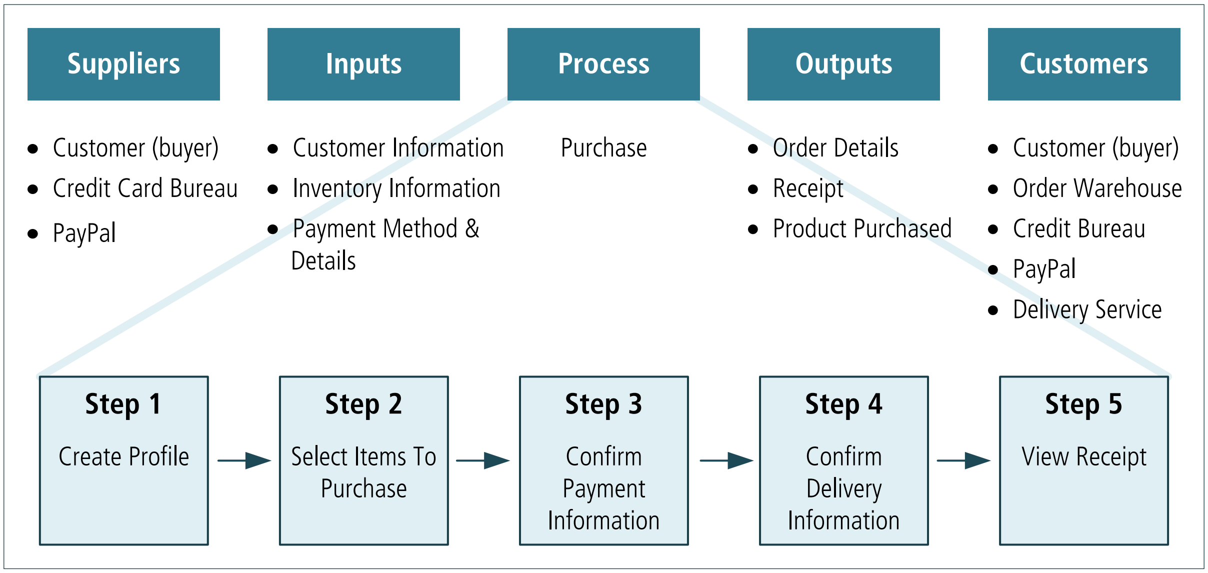
dialogue about problems, opportunities, gaps, root cause, and options and alternatives during process analysis.

Figure 10.34.1: SIPOC Model



Value Stream Mapping (VSM)

Value stream mapping (VSM) is a process analysis method used in Lean methodologies.

Value stream mapping involves the diagramming and monitoring of inputs and application points for processing those inputs, starting from the front-end of the supply chain. At each stage, the value stream map gauges the wait time for the inputs and the actual processing times at the application points (also known as conversion times). At the end of the supply chain, the value stream map depicts the logistics or distribution process to the customer.

The value stream map provides a one-page picture of all the steps involved in the end-to-end process, including both value-adding (the value stream) and non-value-adding (waste) elements.

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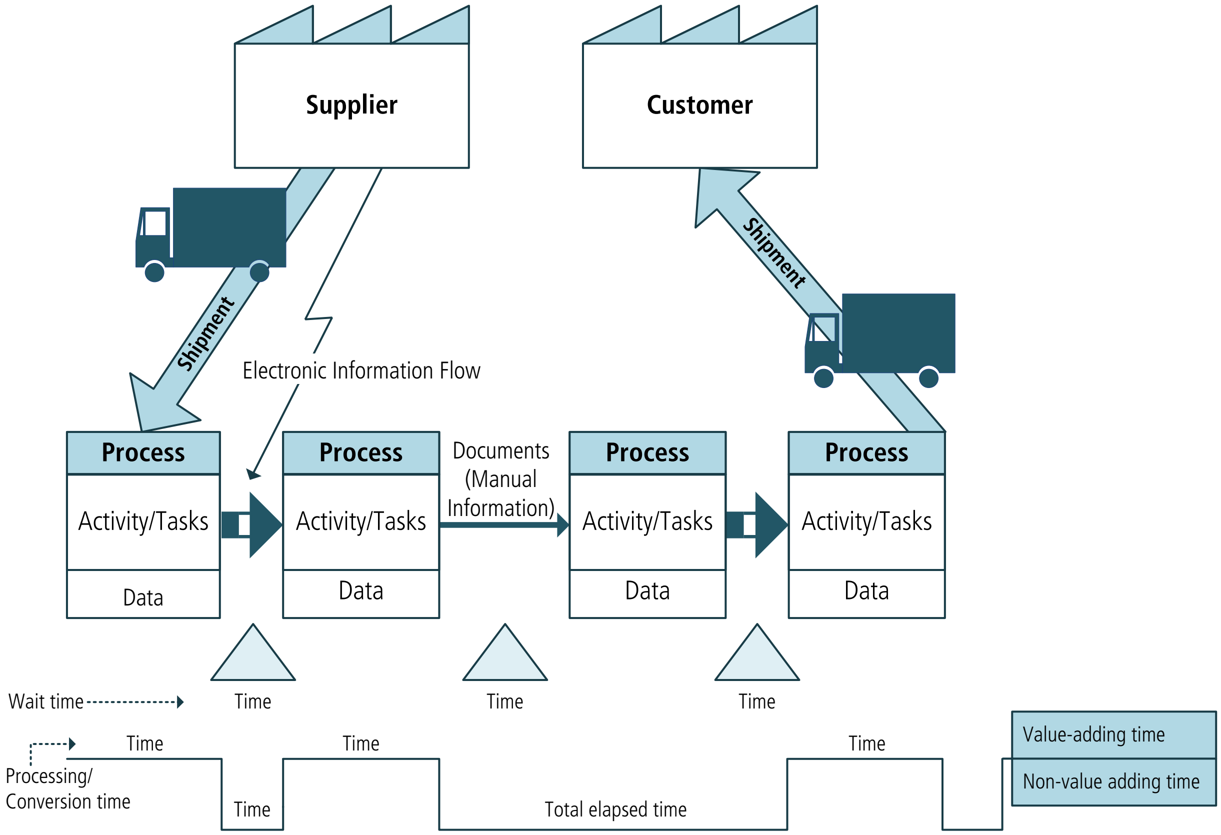
Figure 10.34.2: Value Stream Map



10.34.4 Usage Considerations

.1 Strengths

- Ensures solutions address the right issues, minimizing waste.
- Many different techniques and methodologies can be used and provide teams with great flexibility in approach.

.2 Limitations

- Can be time-consuming.
- There are many techniques and methodologies in process analysis. It can be challenging to decipher which to use and how rigorously to follow them, given the scope and purpose.
- May prove ineffective at process improvement in knowledge or decision-intensive processes.

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10.35 Process Modelling

10.35.1 Purpose

Process modelling is a standardized graphical model used to show how work is carried out and is a foundation for process analysis.

10.35.2 Description

Process models describe the sequential flow of work or activities. A business process model describes the sequential flow of work across defined tasks and activities through an enterprise or part of an enterprise. A system process model defines the sequential flow of control among programs or units within a computer system. A program process flow shows the sequential execution of program statements within a software program. A process model can also be used in documenting operational procedures.

A process model can be constructed on multiple levels, each of which can be aligned to different stakeholder points of view. These levels exist to progressively decompose a complex process into component processes, with each level providing increasing detail and precision. At a high (enterprise or context) level, the model provides a general understanding of a process and its relationship to other processes. At lower (operational) levels, it can define more granular activities and identify all outcomes, including exceptions and alternative paths. At the lowest (system) level, the model can be used as a basis for simulation or execution.

Process models can be used to:

- describe the context of the solution or part of the solution,
- describe what actually happens, or is desired to happen, during a process,
- provide an understandable description of a sequence of activities to an external observer,
- provide a visual to accompany a text description, and
- provide a basis for process analysis.

The business analyst can use a process model to define the current state of a process (also known as an as-is model) or a potential future state (also known as a to-be model). A model of the current state can provide understanding and agreement as to what happens now. A model of the future state can provide alignment with what is desired to happen in the future.

Process models generally include:

- the participants in the process,
- the business event that triggers the process,
- the steps or activities of the process (both manual and automated),
- the paths (flows) and decision points that logically link those activities, and

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- the results of the process.

The most basic process model includes: a trigger event, a sequence of activities, and a result.

A more comprehensive process model can include other elements, such as data/materials, inputs and outputs, and call-out descriptions that supplement the graphical representation.

10.35.3 Elements

.1 Types of Process Models and Notations

Many different notations are used in process modelling.

The most commonly used notations include the following:

- **Flowcharts and Value Stream Mapping (VSM)**: used in the business domain.
- **Data Flow diagrams and Unified Modelling Language™ (UML®) diagrams**: used in the information technology domain.
- **Business Process Model and Notation (BPMN)**: used across both business and information technology domains; is increasingly adopted as an industry standard.
- **Integrated DEFinition (IDEF) notation and Input, Guide, Output, Enabler (IGOE) diagrams**: used for establishing scope.
- **SIPOC and Value Stream Analysis**: used for process modelling.

Process models typically contain some or all of the following key elements:

- **Activity**: an individual step or piece of work that forms part of the business process. It may be a single task or may be further decomposed into a sub-process (with its own activities, flow, and other process elements).
- **Event**: a zero-time occurrence which initiates, interrupts, or terminates an activity or task within a process or the process itself. It may be a message received, the passage of time, or the occurrence of a condition as defined in the business rules.
- **Directional Flow**: a path that indicates the logical sequence of the workflow. In general, diagrams are drawn to show the passage of time in a consistent fashion (typically in the direction that text would be read).
- **Decision Point**: a point in the process where the flow of work splits into two or more flows (paths), which may be mutually exclusive alternatives or parallels. A decision can also be used to locate rules where separate flows merge together.
- **Link**: a connection to other process maps.
- **Role**: a type of person or group involved in the process. Its definitions typically match those in the organizational model.

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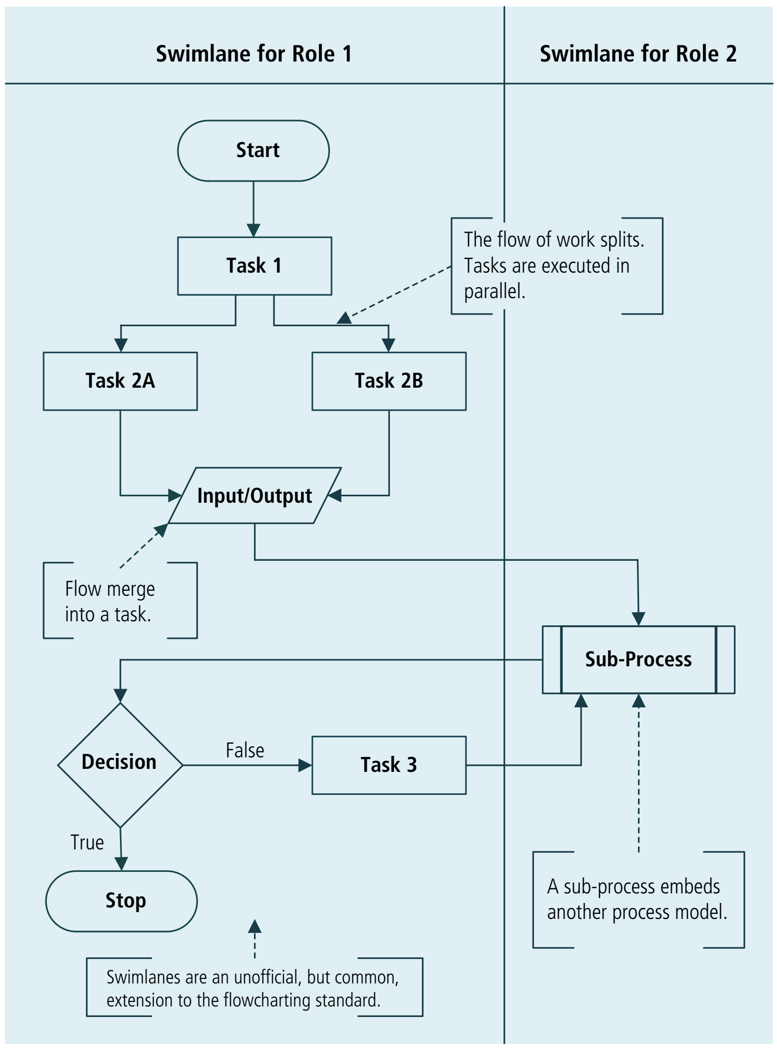
Flowchart

Flowcharts are used commonly with non-technical audiences and are good for gaining both alignment with what the process is and context for a solution. A flowchart can be simple, displaying just the sequence of activities, or it can be more comprehensive, using swimlanes. A swimlane is a partitioned area (horizontal or vertical) that segregates those activities in the process that are carried out by a particular role.

Figure 10.35.1: Flowchart



Business Process Model and Notation (BPMN)

Business Process Model and Notation (BPMN) provides an industry-standard language for modelling business processes in a form that is accessible by both business users and technical developers. BPMN is designed to cover many types of modelling, including both internal (private) processes and collaborative (public) processes. It can be the input to process automation technologies.

A key feature of BPMN is its ability to distinguish the activities of different participants in a process with pools and swimlanes. When the flow of work crosses the boundary of a swimlane, responsibility for the work then passes to

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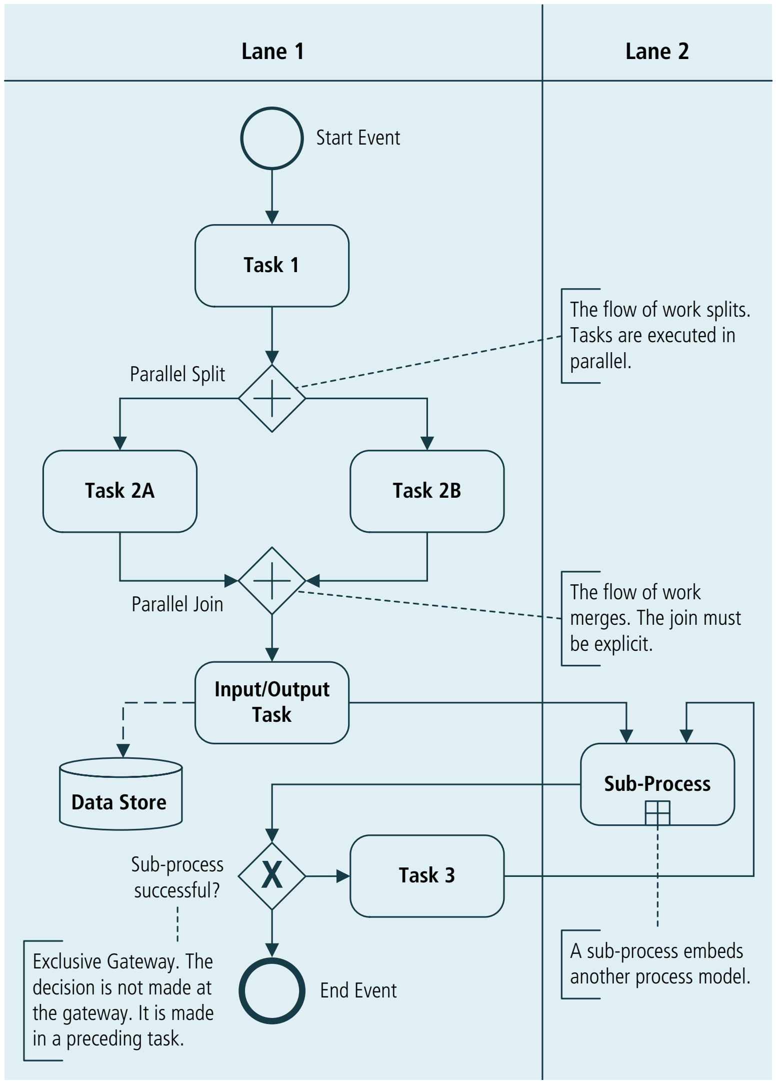
another role within the organization. Swimlanes are part of a pool. A pool is a self-regulating (free-standing) business entity, typically an organization or a system. A pool may include a number of swimlanes, each of which represents a role. Commonly, a process includes one pool for the customer and a second pool for the organization under study, although it is possible for a process to include any number of pools.

Figure 10.35.2: Business Process Model and Notation



Activity Diagram

The activity diagram is one of the use case realization diagrams defined in the Unified Modelling Language™ (UML®). Originally designed to elaborate on a single use case, the activity diagram has been adopted for more general process modelling purposes, including business process modelling. While similar in appearance to a flowchart, the activity diagram typically employs swimlanes to show responsibilities, synchronization bars to show parallel processing, and multiple exit decision points.

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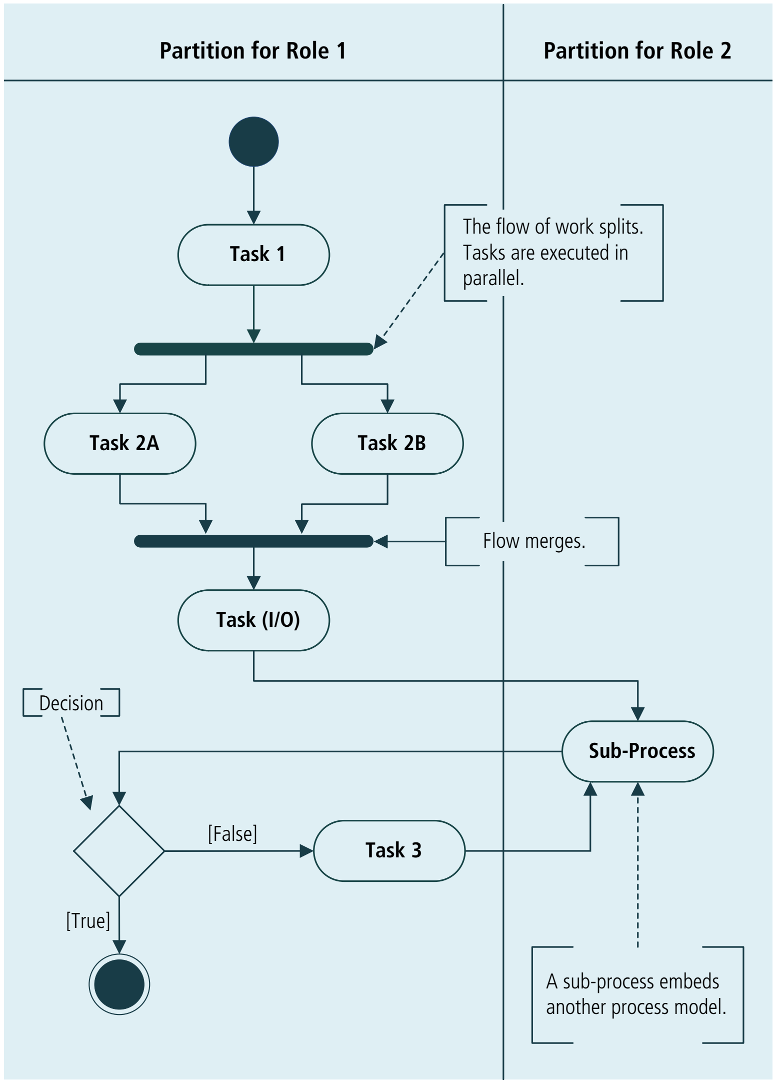
Figure 10.35.3: Activity Diagram



10.35.4 Usage Considerations

.1 Strengths

- Appeals to the basic human understanding of sequential activities.
- Most stakeholders are comfortable with the concepts and basic elements of a process model.
- The use of levels can accommodate the different perspectives of various stakeholder groups.
- Effective at showing how to handle a large number of scenarios and parallel branches.
- Can help identify any stakeholder groups that may have otherwise been overlooked.
- Facilitates the identification of potential improvements by highlighting “pain points” in the process structure (i.e. process visualization).

- Likely to have value in its own right. They provide documentation for compliance purposes and can be used by business stakeholders for training and coordination of activities.
- Can be used as a baseline for continuous improvement.
- Ensures labelling consistency across artifacts.
- Provides transparency and clarity to process owners and participants on activity responsibilities, sequence and hand-overs.

.2 Limitations

- To many people in IT, a formal process model tends to reflect an older and more document-heavy approach to software development. Therefore, project time is not allocated to developing a process model, especially of the current state or problem domain.
- Can become extremely complex and unwieldy if not structured carefully. This is especially true if business rules and decisions are not managed separately from the process.
- Complex processes can involve many activities and roles; this can make them almost impossible for a single individual to understand and 'sign off'.
- Problems in a process cannot always be identified by looking at a high-level model. A more detailed model with reference to metadata (such as path frequency, cost, and time factors) is usually required. It is often necessary to engage with stakeholders directly to find the operational problems they have encountered while working with a process.
- In a highly dynamic environment where things change quickly, process models can become obsolete.
- May prove difficult to maintain if the process model only serves as documentation, as stakeholders may alter the process to meet their needs without updating the model.

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10.36 Prototyping

10.36.1 Purpose

Prototyping is used to elicit and validate stakeholder needs through an iterative process that creates a model or design of requirements. It is also used to optimize user experience, to evaluate design options, and as a basis for development of the final business solution.

10.36.2 Description

Prototyping is a proven method for product design. It works by providing an early model of the final result, known as a prototype. Prototyping is used to identify both missing or improperly specified requirements and unsubstantiated assumptions by demonstrating what the product looks like and how it acts in the early stages of design.

Prototypes can be non-working models, working representations, or digital depictions of a solution or a proposed product. They can be used to mock up websites, serve as a partially working construct of the product, or describe processes through a series of diagrams (such as workflow). Business rules and data prototypes can be used to discover desired process flow and business rules. Data prototyping can be used for data cleansing and transformation.

10.36.3 Elements

.1 Prototyping Approach

There are two common approaches to prototyping:

- **Throw-away**: prototypes are generated with simple tools (such as paper and pencil, a whiteboard, or software) to serve the goal of uncovering and clarifying requirements. The prototype may be updated or evolve during the course of discussion and development, but does not become workable code or get maintained as a deliverable once the final system or process is implemented. This method is helpful for identifying functionality or processes that are not easily elicited by other techniques, have conflicting points of view, or are difficult to understand. These prototypes can be an inexpensive tool to uncover or confirm requirements that go beyond an interface including requirements related to processes, data, and business rules.
- **Evolutionary or Functional**: prototypes are created to extend initial requirements into a functioning solution as requirements are further defined through stakeholder use. This approach produces a working solution and usually requires a specialized prototyping tool or language. These prototypes may be used in the final solution. If specialized software is used, business processes, rules, and data can be simulated to evaluate the impact of changes and validate desired outcomes.

.2 Prototyping Examples

There are many forms of prototyping in use today.

Each of the following can be considered a form of prototyping:

- **Proof of Principle or Proof of Concept**: is a model created to validate the design of a system without modelling the appearance, materials used in the creation of work, or processes/workflows ultimately used by the stakeholders.
- **Form Study Prototype**: is used to explore the basic size, look, and feel of a product that will be manufactured, without creating actual functionality. It is used to assess ergonomic and visual factors using a sculptural representation of the product made from inexpensive materials. This type of prototype may also be used to model a workflow or navigation at a high

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level in order to identify gaps or inconsistencies in the possible solution of the properties (for example, appearance, configuration).

- **Usability Prototype**: is a product model created to test how the end user interacts with the system without including any of the properties (for example, appearance, configuration).
- **Visual Prototype**: is a product model created to test the visual aspects of the solution without modelling the complete functionality.
- **Functional Prototype**: is a model created to test software functionality, qualities of the system for the user (for example, appearance), and workflow. It is also referred to as a working model and is used both to simulate business processes and business rules and to evaluate software function calls.

.3 Prototyping Methods

The following is a list of commonly used methods for prototyping:

- **Storyboarding**: is used to visually and textually detail the sequence of activities by summing up different user interactions with the solution or enterprise.
- **Paper Prototyping**: uses paper and pencil to draft an interface or process.
- **Workflow Modelling**: depicts a sequence of operations that are performed and usually focuses solely on the human aspect.
- **Simulation**: is used to demonstrate solutions or components of a solution. It may test various processes, scenarios, business rules, data, and inputs.

10.36.4 Usage Considerations

.1 Strengths

- Provides a visual representation for the future state.
- Allows for stakeholders to provide input and feedback early in the design process.
- When using throw-away or paper prototyping methods, users may feel more comfortable being critical of the mock-up because it is not polished and release-ready.
- A narrow yet deep vertical prototype can be used for technical feasibility studies, proof of concept efforts, or to uncover technology and process gaps.

.2 Limitations

- If the system or process is highly complex, the prototyping process may become bogged down with discussion of 'how' rather than 'what', which can make the process take considerable time, effort, and facilitation skill.
- Underlying technology may need to be understood or assumed in order to initiate prototyping.

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- If the prototype is deeply elaborate and detailed, stakeholders may develop unrealistic expectations for the final solution. These can range from assumed completion dates to higher expectations of performance, reliability, and usability.
- Stakeholders may focus on the design specifications of the solution rather than the requirements that any solution must address. This can, in turn, constrain the solution design. Developers may believe that they must provide a user interface that precisely matches the prototype, even if more elegant technology and interface approaches exist.

10.37 Reviews

10.37.1 Purpose

Reviews are used to evaluate the content of a work product.

10.37.2 Description

Different types of reviews are conducted for business analysis work products.

Each is tailored to the needs of the organization and business analyst, and uses these dimensions:

- **Objectives**: defining the purpose of the review.
- **Techniques**: identifying either a formal or informal way to perform the review.
- **Participants**: identifying who should take part in the review activity.

Each review is focused on a work product, not the skills or actions of the participants. The work product may be a package of several deliverables, a single deliverable, a portion of a deliverable, or work in process. For a completed work product, the objective of the review is usually to remove defects or inform the reviewers about the content. For work in process, the review may be conducted to resolve an issue or question.

Each review includes the business analyst as a participant. Reviewers may be peers, especially for work in process, or stakeholders, who validate that the work product is complete and correct. The review steps depend on the technique used.

Reviews can include:

- an overview of the work product and review objectives,
- checklists and reference materials that can be used by reviewers,
- reviewing the work product and documenting the findings, and
- verifying any rework.

Using feedback from reviewers, the business analyst updates the work product.

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10.37.3 Elements

.1 Objectives

Objectives are clearly communicated to all participants prior to the review.

Objectives may include one or more goals, for example:

- to remove defects,
- to ensure conformance to specifications or standards,
- to ensure the work product is complete and correct,
- to establish consensus on an approach or solution,
- to answer a question, resolve an issue, or explore alternatives,
- to educate reviewers about the work product, and
- to measure work product quality.

.2 Techniques

Reviews can be formal or informal. The techniques used during a review are selected to support the objectives of the review.

The following techniques are commonly used by business analysts when conducting reviews:

- **Inspection**: a formal technique that includes an overview of the work product, individual review, logging the defects, team consolidation of defects, and follow-up to ensure changes were made. The focus is to remove defects and create a high quality work product. While usually performed by peers, it can also be used for stakeholder reviews.
- **Formal Walkthrough (also known as Team Review)**: a formal technique that uses the individual review and team consolidation activities often seen in inspection. Walkthroughs are used for peer reviews and for stakeholder reviews.
- **Single Issue Review (also known as Technical Review)**: a formal technique focused on either one issue or a standard in which reviewers perform a careful examination of the work product prior to a joint review session held to resolve the matter in focus.
- **Informal Walkthrough**: an informal technique in which the business analyst runs through the work product in its draft state and solicits feedback. Reviewers may do minimal preparation before the joint review session.
- **Desk Check**: an informal technique in which a reviewer who has not been involved in the creation of the work product provides verbal or written feedback.
- **Pass Around**: an informal technique in which multiple reviewers provide verbal or written feedback. The work product may be reviewed in a common copy of the work product or passed from one person to the next.

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- **Ad hoc**: an informal technique in which the business analyst seeks informal review or assistance from a peer.

.3 Participants

Participant roles involved in any particular review depend on the objectives of the review, the selected technique, and any organizational standards that may be in place.

In some situations, a supervisor or manager may be one of the reviewers because of their expertise. In these situations, the moderator is careful to avoid adversely affecting the level of candour of other participants or inappropriately affecting decisions of the team.

Table 10.37.1: Review Roles

Role	Description	Responsibility	Applicable Techniques
Author	Author of the work product.	Answers questions about the work product and listens to suggestions and comments. Incorporates changes into the work product after the review.	All
Reviewer	A peer or stakeholder.	Examines the work product according to the review objectives. For defect detection reviews, the reviewer examines the work product prior to a review session and keeps track of both defects found and suggestions for improvement.	All
Facilitator	A neutral facilitator (should not be the author in order to avoid compromising the review).	Facilitates the review session, keeps participants focused on the objectives of the review and ensures that each relevant section of the work product is covered. Validates that reviewers have examined the work product before the session begins and ensures that all reviewers participate in the review session.	• Inspection • Formal walkthrough • May be helpful for single issue review
Scribe	A neutral participant with strong communication skills.	Documents all defects, suggestions, comments, issues, concerns, and outstanding questions that are raised during a review session. Familiarity with the subject matter enables the scribe to capture items clearly.	• Inspection • Formal and informal walkthrough

10.37.4 Usage Considerations

.1 Strengths

- Can help identify defects early in the work product life cycle, eliminating the need for expensive removal of defects discovered later in the life cycle.
- All parties involved in a review become engaged with the final outcome; they have a vested interest in a quality result.

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- Desk checks and pass around reviews can be performed by a reviewer at a convenient time, rather than interrupting work in progress to attend a meeting.

.2 Limitations

- Rigorous team reviews take time and effort. Thus, only the most critical work products might be reviewed using inspection or formal walkthrough techniques.
- Informal reviews by one or two reviewers are practical in terms of the effort required, but they provide less assurance of removing all significant defects than using a larger team and more formal process.
- For desk checks and pass around reviews it may be difficult for the author to validate that an independent review was done by each participant.
- If review comments are shared and discussed via e-mail there may be many messages to process, which makes it difficult for the author to resolve disagreements or differences in suggested changes.

10.38 Risk Analysis and Management

10.38.1 Purpose

Risk analysis and management identifies areas of uncertainty that could negatively affect value, analyzes and evaluates those uncertainties, and develops and manages ways of dealing with the risks.

10.38.2 Description

Failure to identify and manage risks may negatively affect the value of the solution. Risk analysis and management involves identifying, analyzing, and evaluating risks. Where sufficient controls are not already in place, business analysts develop plans for avoiding, reducing, or modifying the risks, and when necessary, implementing these plans.

Risk management is an ongoing activity. Continuous consultation and communication with stakeholders helps to both identify new risks and to monitor identified risks.

10.38.3 Elements

.1 Risk Identification

Risks are discovered and identified through a combination of expert judgment, stakeholder input, experimentation, past experiences, and historical analysis of similar initiatives and situations. The goal is to identify a comprehensive set of relevant risks and to minimize the unknowns. Risk identification is an ongoing activity.

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A risk event could be one occurrence, several occurrences, or even a non-occurrence. A risk condition could be one condition or a combination of conditions. One event or condition may have several consequences, and one consequence may be caused by several different events or conditions.

Each risk can be described in a risk register that supports the analysis of those risks and plans for addressing them.

Figure 10.38.1: Example of a Risk Register

#	Risk Event or Condition	Consequence	Probability	Impact	Risk Level	Risk Modification Plan	Risk Owner	Residual Risk		
								Probability	Impact	Risk Level
1	If the union does not agree with changes to job descriptions	then planned staff changes will not be able to occur	Medium	Medium	Medium	Begin consultations with the union no later than next month	Marta	Low	Low	Low
2	If subject matter experts are not available for requirements elicitation	then scope and quality will be reduced, and the delivery date will be pushed back	Medium	High	High	Develop a plan for when the SME's are required, hold on-site workshops and obtain agreement from the sponsor about their participation	Deepak	Low	Medium	Low
3	If an insufficient number of customers reply to our survey	then we will not have a representative sample of customer requirements	Medium	High	High	Contract with a firm that specializes in survey management to develop and run the survey	François	Low	Medium	Low
4	If the organizational structure does not adjust to the new business processes	then the enterprise will not be able to achieve the planned efficiencies and the business need will not be met	High	High	High	The business sponsor must approve the organizational changes prior to deployment, and the changes must occur prior to deployment	Jiahui	Medium	Low	Medium

.2 Analysis

Analysis of a risk involves understanding the risk, and estimating the level of a risk. Sometimes controls may already be in place to deal with some risks, and these should be taken into account when analyzing the risk.

The likelihood of occurrence could be expressed either as a probability on a numerical scale or with values such as Low, Medium, and High.

The consequences of a risk are described in terms of their impact on the potential value. The impact of any risk can be described in terms of cost, duration, solution scope, solution quality, or any other factor agreed to by the stakeholders such as reputation, compliance, or social responsibility.

Table 10.38.1: Example of a Risk Impact Scale

	Scope	Quality	Cost	Effort	Duration	Reputation	Social Responsibility
Low Impact	Minor areas of scope are affected	Minor quality problems	Less than 1% cost impact	Less than 2% extra days effort	Delay of up to 3%	Very minor impact to enterprise's reputation	Minor impediment
Medium Impact	Major areas of scope are affected, but workarounds are feasible	Significant quality issues, but the product is still usable	More than 1% but less than 3% impact	2% -10% extra days effort	Delay of 3%-10%	Moderate impact to enterprise's reputation	Major impediment
High Impact	The product does not meet the business need	The product is not usable	More than 3% impact	More than 10% extra days effort	Delay of more than 10%	Severe impact to enterprise's reputation	Severe impediment

While an enterprise may have a standard or baseline risk impact scale, the categories like cost, effort, and reputation, and the thresholds may be adjusted to consider the potential value and the level of risk that is acceptable. Typically, three to five broad categories of level are used to describe how to interpret the potential impact.

The level of a given risk may be expressed as a function of the probability of occurrence and the impact. In many cases, it is a simple multiplication of probability and impact. The risks are prioritized relative to each other according to their level. Risks which could occur in the near term may be given a higher priority than risks which are expected to occur later. Risks in some categories such as reputation or compliance may be given higher priority than others.

.3 Evaluation

The risk analysis results are compared with the potential value of the change or of the solution to determine if the level of risk is acceptable or not. An overall risk level may be determined by adding up all the individual risk levels.

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.4 Treatment

Some risks may be acceptable, but for other risks it may be necessary to take measures to reduce the risk.

One or more approaches for dealing with a risk may be considered, and any combination of approaches could be used to address a risk:

- **Avoid**: either the source of the risk is removed, or plans are adjusted to ensure that the risk does not occur.
- **Transfer**: the liability for dealing with the risk is moved to, or shared with, a third party.
- **Mitigate**: reduce the probability of the risk occurring or the possible negative consequences if the risk does occur.
- **Accept**: decide not to do anything about the risk. If the risk does occur, a workaround will be developed at that time.
- **Increase**: decide to take on more risk to pursue an opportunity.

Once the approach for dealing with a specific risk is selected, a risk response plan is developed and assigned to a risk owner with responsibility and authority for that risk. In the case of risk avoidance, the risk owner takes steps to ensure that the probability or the impact of the risk is reduced to nil. For those risks which cannot be reduced to nil, the risk owner is responsible for monitoring the risk, and for implementing a risk mitigation plan.

The risk is re-analyzed to determine the residual risk which is the new probability and new impact as a result of the measures taken to modify the risk. There could be a cost-benefit analysis done to determine if the cost and effort of the measures reduces the level of risk enough to make it worthwhile. The risks may be re-evaluated in terms of the residual risk.

Stakeholders should be informed of the plans for modifying the risks.

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10.38.4 Usage Considerations

.1 Strengths

- Can be applied to strategic risks which affect long-term value of the enterprise, tactical risks which affect the value of a change, and operational risks which affect the value of a solution once the change is made.
- An organization typically faces similar challenges on many of its initiatives. The successful risk responses on one initiative can be useful lessons learned for other initiatives.
- The risk level of a change or of a solution could vary over time. Ongoing risk management helps to recognize that variation, and to re-evaluate the risks and the suitability of the planned responses.

.2 Limitations

- The number of possible risks to most initiatives can easily become unmanageably large. It may only be possible to manage a subset of potential risks.
- There is the possibility that significant risks are not identified.

10.39 Roles and Permissions Matrix

10.39.1 Purpose

A roles and permissions matrix is used to ensure coverage of activities by denoting responsibility, to identify roles, to discover missing roles, and to communicate results of a planned change.

10.39.2 Description

Role and permission allocation involves identifying roles, associating these with solution activities, and then denoting authorities who can perform these activities. A role is a label for a group of individuals who share common functions. Each function is portrayed as one or more solution activities. A single activity can be associated with one or more roles by designating authorities. Each individual that is assigned this authority can perform the associated activity.

The following is an example of a roles and permissions matrix for a software system.

Figure 10.39.1: Roles and Permissions Matrix

Roles and Permissions Matrix Activity	Role Group 1	Administrator	Manager	Role Group 2	Sales	Customer
Create new account		X	X			X
Modify account		X	X			X
Create order		X	X		X	X
View reports		X	X		X	
Create reports		X	X		X	

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10.39.3 Elements

.1 Identifying Roles

To identify roles for either internal or external stakeholders, business analysts:

- review any organizational models, job descriptions, procedure manuals, and system user guides, and
- meet with stakeholders to uncover additional roles.

Through this review and discussion, the business analyst considers both that individuals with the same job title may have different roles and that individuals with different job titles may have the same roles.

When identifying roles, business analysts look for common functions that are performed by individuals with similar needs.

.2 Identifying Activities

Business analysts frequently use functional decomposition to break down each function into sub-parts, process modelling to better understand the workflow and division of work among users, and use cases to represent tasks. By performing these techniques, the business analyst can ensure that all functions are accounted for and their activities are identified among various use case scenarios.

There may be different levels of abstraction for roles and permission matrices based on the business analysis perspective. Initiative level roles and responsibilities may be identified in a RACI (Responsible, Accountable, Consulted, Informed) matrix. Specific information technology system roles and responsibilities may be identified in a CRUD (Create, Read, Update, and Delete) matrix.

.3 Identifying Authorities

Authorities are actions that identified roles are permitted to perform. For each activity, the business analyst identifies the authorities for each role. When identifying authorities, business analysts consider the level of security needed and how the work flows through the process. Business analysts collaborate with stakeholders to validate identified authorities.

.4 Refinements

Delegations

The business analyst may also identify which authorities can be delegated by one individual to another on a short-term or permanent basis.

Inheritances

Stakeholders may request that when an individual is assigned an authority at an organizational hierarchy level that this assignment pertain to only that user's organizational level and any subsidiary organizational unit levels.

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10.39.4 Usage Considerations

.1 Strengths

- Provides procedural checks and balances, as well as data security, by restricting individuals from performing certain actions.
- Promotes improved review of transaction history, in that audit logs can capture details about any assigned authorities at the time.
- Provides documented roles and responsibilities for activities.

.2 Limitations

- Need to recognize the required level of detail for a specific initiative or activity; too much detail can be time-consuming and not provide value, too little detail can exclude necessary roles or responsibilities.

10.40 Root Cause Analysis

10.40.1 Purpose

Root cause analysis is used to identify and evaluate the underlying causes of a problem.

10.40.2 Description

Root cause analysis is a systematic examination of a problem or situation that focuses on the problem's origin as the proper point of correction rather than dealing only with its effects. It applies an iterative analysis approach in order to take into account that there might be more than one root cause contributing to the effects. Root cause analysis looks at the main types of causes such as people (human error, lack of training), physical (equipment failure, poor facility), or organizational (faulty process design, poor structure).

Root cause analysis helps organize the information in a framework, which allows for deeper analysis if needed. Root cause analysis can be used for:

- **Reactive Analysis**: identifying the root cause(s) of an occurring problem for corrective action, or
- **Proactive Analysis**: identifying potential problem areas for preventive action.

Root cause analysis uses four main activities:

- **Problem Statement Definition**: describes the issue to be addressed.
- **Data Collection**: gathers information about the nature, magnitude, location, and timing of the effect.
- **Cause Identification**: investigates the patterns of effects to discover the specific actions that contribute to the problem.

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- **Action Identification**: defines the corrective action that will prevent or minimize recurrence.

10.40.3 Elements

.1 The Fishbone Diagram

A fishbone diagram (also known as an Ishikawa or cause-and-effect diagram) is used to identify and organize the possible causes of a problem. This tool helps to focus on the cause of the problem versus the solution and organizes ideas for further analysis. The diagram serves as a map that depicts possible cause-and-effect relationships.

Steps to develop a fishbone diagram include:

Step 1. Capturing the issue or problem under discussion in a box at the top of the diagram.

Step 2. Drawing a line from the box across the paper or whiteboard (forming the spine of the fishbone).

Step 3. Drawing diagonal lines from the spine to represent categories of potential causes of the problem. The categories may include people, processes, tools, and policies.

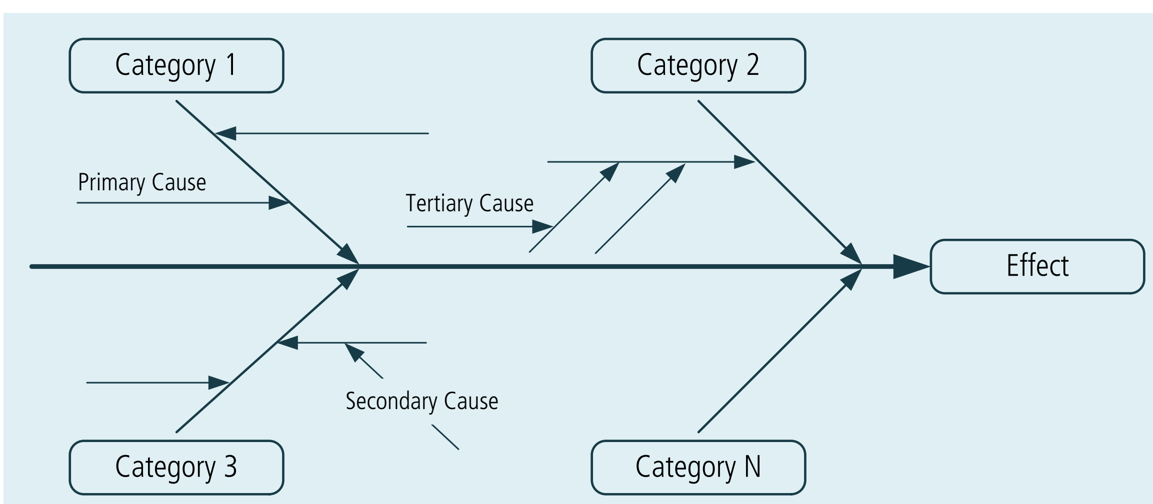
Step 4. Drawing smaller lines to represent deeper causes.

Step 5. Brainstorming categories and potential causes of the problem and capturing them under the appropriate category.

Step 6. Analyzing the results. Remember that the group has identified only potential causes of the problem. Further analysis is needed to validate the actual cause, ideally with data.

Step 7. Brainstorming potential solutions once the actual cause has been identified.

Figure 10.40.1: Fishbone Diagram



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.2 The Five Whys

The five whys is a question asking process to explore the nature and cause of a problem. The five whys approach repeatedly asks questions in an attempt to get to the root cause of the problem. This is one of the simplest facilitation tools to use when problems have a human interaction component.

To use this technique:

Step 1. Write the problem on a flip chart or whiteboard.

Step 2. Ask "Why do you think this problem occurs?" and capture the idea below the problem.

Step 3. Ask "Why?" again and capture that idea below the first idea.

Continue with step 3 until you are convinced the actual root cause has been identified. This may take more or less than five questions—the technique is called the five whys because it often takes that many to reach the root cause, not because the question must be asked five times.

The five whys can be used alone or as part of the fishbone diagram technique. Once all ideas are captured in the diagram, use the five whys approach to drill down to the root causes.

10.40.4 Usage Considerations

.1 Strengths

- Helps to maintain an objective perspective when performing cause-and-effect analysis.
- Enables stakeholders to specify an effective solution at the appropriate points for corrective action.

.2 Limitations

- Works best when the business analyst has formal training to ensure the root causes, not just symptoms of the problem, are identified.
- May be difficult with complex problems; the potential exists to lead to a false trail and/or dead–end conclusion.

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